

T0-Time-Mass-Duality Theory: Compelling Derivation of Fractal Dimension D_f from Lepton Mass Ratio

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Abstract

The T0-Time-Mass-Duality theory derives fundamental constants and masses parameter-free from the universal geometric parameter $\xi = 4/30000$. This complementary document examines a validation of the fractal dimension $D_f = 3 - \xi \approx 2.99987$ through backward derivation from the experimental mass ratio $r = m_\mu/m_e \approx 206.768$ (CODATA 2025). The direct geometric calculation chain used here does not hold in its present form: the electron formula gives 0.0854 GeV instead of m_e , the muon chain gives m_e instead of m_μ , and the Nambu inversion is not consistent with the torsion mass m_T ; a validation of D_f does not follow from it. The systematic mass calculation is presented in *006_T0_Teilchenmassen_En.pdf*.

.1 Introduction

Important

Document Complementarity This document focuses on **examining a validation of the fractal dimension D_f** from experimental lepton masses. It complements the main document *006_T0_Teilchenmassen_En.pdf*, which presents the complete systematic mass calculation for all fermions.

Particle physics faces the fundamental problem of arbitrary mass parameters in the Standard Model. The T0-Time-Mass-Duality theory replaces this approach with a completely parameter-free description.

.2 Parameters and Basic Formulas

The theory is based on time-energy duality and fractal spacetime structure.

Exact Geometric Parameters

$$\xi = \frac{4}{30000} = \frac{1}{7500} \approx 1.333 \times 10^{-4}, \quad (1)$$

$$D_f = 3 - \xi \approx 2.99986667, \quad (2)$$

$$\alpha = \frac{1 - \xi}{137} \approx 7.298 \times 10^{-3}, \quad (3)$$

$$K_{\text{frac}} = 1 - 100\xi \approx 0.9867, \quad (4)$$

$$g_{T0}^2 = \alpha K_{\text{frac}}, \quad (5)$$

$$E_0 = \frac{1}{\xi} \approx 7500 \text{ GeV}, \quad (6)$$

$$p = -\frac{2}{3}. \quad (7)$$

$1/\xi = 7500$ is a pure number; assigning the unit GeV in Eq. (6) is a setting, not a derivation (other documents read the same number as eV). In T0 units with $\alpha = 1$ one has $\xi E_0^2 = 1$, i.e. $E_0^2 = 1/\xi = 7500$; the number belongs to E_0^2 , not to E_0 (Doc. 386). The corpus value is $E_0 = \sqrt{m_e m_\mu / K_{\text{frac}}} = 7.398 \text{ MeV}$; 7500 GeV exceeds it by a factor of 1.01×10^6 . The following formulas use $E_0 \xi = 1 \text{ GeV}$ and depend on this setting.

Fine Structure Constant Precision The deviation of α from CODATA is only $\approx 0.013\%$ – strong evidence for the fractal correction.

.3 Geometric Mass Derivation - Direct Method

T0-theory offers several mathematically equivalent methods for mass calculation. In this document the **direct geometric method** is examined as to whether it validates the fractal dimension.

Electron Mass m_e - Direct Geometric Method

In the direct geometric method:

$$m_e = E_0 \cdot \xi \cdot \sqrt{\alpha} \cdot \frac{\Gamma(D_f)}{\Gamma(3)}. \quad (8)$$

With the parameters of Section .2 ($E_0 \xi = 1 \text{ GeV}$, $\sqrt{\alpha} = 0.08543$, $\Gamma(2.99987)/\Gamma(3) = 0.99994$) the formula gives $m_e = 0.0854 \text{ GeV}$ – larger than the target value $5.10 \times 10^{-4} \text{ GeV}$ by a factor of 167. The formula does not yield m_e in this form. The deviation of -0.20% from CODATA (0.000,511 GeV) and the following table refer to the target value, not to the result of the formula.

Consistency Check with Main Document

Method	m_e [GeV]	Accuracy	Source
Direct geometric (target value)	5.10×10^{-4}	99.8%	This document
Extended Yukawa	5.11×10^{-4}	99.9%	006
Experiment (CODATA)	5.11×10^{-4}	100%	Reference

Table 1: Consistency of mass calculation methods in T0-theory

Method Comparison The agreement within 0.2% holds for the target value of the direct method, not for the result of Eq. (8); an equivalence of the methods is therefore not shown. The Yukawa method bridges to the Standard Model.

Effective Torsion Mass m_T

$$R_f = \frac{\Gamma(D_f)}{\Gamma(3)} \sqrt{\frac{E_0}{m_e}}, \quad (9)$$

$$m_T = \frac{m_e}{\xi} \sin(\pi\xi) \pi^2 \sqrt{\frac{\alpha}{K_{\text{frac}}}} R_f \approx 5.220 \text{ GeV}. \quad (10)$$

Muon Mass m_μ

From RG-duality and loop integral I :

$$I = \int_0^1 \frac{m_e^2 x(1-x)^2}{m_e^2 x^2 + m_T^2(1-x)} dx \approx 1.60 \times 10^{-9}, \quad (11)$$

$$r \approx \sqrt{6I} \approx 9.8 \times 10^{-5}, \quad (12)$$

$$m_T \cdot r \approx 5.11 \times 10^{-4} \text{ GeV}. \quad (13)$$

The integral is evaluated numerically with $m_e = 5.11 \times 10^{-4} \text{ GeV}$ and $m_T = 5.220 \text{ GeV}$. The product $m_T r$ thus gives m_e , not m_μ (0.105,658 GeV); the chain does not yield m_μ in this form.

Important

Mass Ratio Since the chain does not yield m_μ , no mass ratio $r = m_\mu/m_e$ can be calculated from it; an independent validation of the geometric foundation does not result here.

.4 Backward Validation: D_f from r and Nambu Formula

The classical Nambu formula $r \approx (3/2)/\alpha$ (dev. -0.58%) is refined by the ξ -correction.

Nambu Inversion

$$m_T^{\text{target}} = \frac{m_\mu}{\sqrt{\alpha} \cdot (3/2) \cdot (1 - \xi)} = \frac{0.10566}{0.08543 \times 1.5 \times 0.99987} \approx 0.825 \text{ GeV}. \quad (14)$$

This value is not consistent with Equation (10) ($m_T = 5.220 \text{ GeV}$); the inversion does not hold in this form.

Optimization for D_f

Define $m_T(D_f)$ according to Equation 10 and solve:

$$D_f = \arg \min |m_T(D_f) - m_T^{\text{target}}|. \quad (15)$$

Key Result

Result of the Backward Validation Since $m_T^{\text{target}} = 0.825$ GeV and $m_T = 5.220$ GeV do not agree, the optimization does not hold on this basis; a result $D_f \approx 2.99986667$ is not supported by this calculation. That the experimental mass ratio compels the fractal geometry does not follow from this calculation; the foundations of *006_T0_Teilchenmassen_En.pdf* are not independently validated here.

.5 Application: Anomalous Magnetic Moment a_μ^{T0}

With the fractal dimension $D_f = 3 - \xi$ and geometric masses:

$$F_2^{\text{T0}}(0) = \frac{g_{T0}^2}{8\pi^2} I_\mu K_{\text{frac}}, \quad (16)$$

$$\text{term} = \left(\frac{\xi E_0}{m_T} \right)^p = m_T^{2/3}, \quad (17)$$

$$F_{\text{dual}} = \frac{1}{1 + \text{term}} \approx 0.249, \quad (18)$$

$$a_\mu^{\text{T0}} = F_2^{\text{T0}}(0) \cdot F_{\text{dual}} \approx 1.53 \times 10^{-9} = 153 \times 10^{-11}. \quad (19)$$

Experimental Validation Status 2025 (Fermilab final result, 2025 White Paper of the Muon $g-2$ Theory Initiative): $\Delta a_\mu = (38 \pm 63) \times 10^{-11}$. The T0 value 153×10^{-11} lies $(153-38)/63 \approx 1.8\sigma$ above it.

.6 Python Implementation and Reproducibility

Important

Full Transparency For reproduction of all numerical calculations see the external script `t0_df_from_masses_geometry.py` in the repository folder.

.7 References

- Pascher, J. (2025). *T0-Model: Complete Parameter-Free Particle Mass Calculation* (006_T0_Teilchenmassen_En.pdf). Available at:
- Pascher, J. (2025). *T0-Time-Mass-Duality Repository*, GitHub v1.6. Available at:
- CODATA (2025). *Fundamental Physical Constants*, NIST.